In [ ]:
```
import os
os.chdir(r"c:\users\ADMIN")
```

In [ ]:
```
os.mkdir("2601008_Prasad_Bairagi")
os.chdir("2601008_Prasad_Bairagi")
os.getcwd()
```

In [77]:
```
# 1. Read a file containing student names (one per line) and store them in a lis
file = open("students1.txt", "r")
names = file.read().split()
file.close()
print(names)
```

```
['Prasad', 'Pranav', 'Rudraksh', 'Aayush', 'Gaurav', 'Shrushti', 'Rohan', 'Prach
i', 'OM']
```

In [78]:
```
#2. Print names longer than 5 characters.
for name in names:
    if len(name) > 5:
        print(name)
```

```
Prasad
Pranav
Rudraksh
Aayush
Gaurav
Shrushti
Prachi
```

In [16]:
```
#3. Count how many names start with a vowel.
count = 0
for name in names:
    if name[0].lower() in "aeiou":
        count += 1
print(count)
```

```
2
```

In [80]:
```
#4. Save the filtered names (longer than 5 characters) into a new file.
names = ["Rohan", "Pooja", "Vikramaditya", "Ananya", "Raj"]
with open("filtered_names.txt", "w") as file:
    for name in names:
        clean_name = name.strip()
        if len(clean_name) > 5:
            file.write(clean_name)
```

In [30]:
```
#5. Sort the list of names alphabetically and print them in ascending order.
sorted_list = sorted(names)
print(sorted_list)
```

```
['Aayush', 'Gaurav', 'OM', 'Prachi', 'Pranav', 'Prasad', 'Rohan', 'Rudraksh', 'Sh
rushti']
```

In [31]:
```
#6. Print names that contain the letter "a".
for name in names:
    if "a" in name.lower():
        print(name)
```

```
Prasad
Pranav
Rudraksh
Aayush
Gaurav
Rohan
Prachi
```

In [37]:
```
#7. Count word frequency in names (handle duplicates) and store results in a dic
from collections import Counter
frequency = Counter(name.lower() for name in names)
print(dict(frequency))
```

```
{'prasad': 1, 'pranav': 1, 'rudraksh': 1, 'aayush': 1, 'gaurav': 1, 'shrushti':
1, 'rohan': 1, 'prachi': 1, 'om': 1}
```

In [38]:
```
#8. String Analysis
for name in names:
    clean = name.replace(" ", "").lower()
    print(name, len(name))
    print(name[0], name[-1])
    print(clean == clean[::-1])
```

```
Prasad 6
P d
False
Pranav 6
P v
False
Rudraksh 8
R h
False
Aayush 6
A h
False
Gaurav 6
G v
False
Shrushti 8
S i
False
Rohan 5
R n
False
Prachi 6
P i
False
OM 2
O M
False
```

In [79]:
```
#9. File + List + String Combined Challenge

file = open("students1.txt", "r")
names = [line.strip() for line in file.readlines()]
unique_names = list(set(names))
unique_names.sort(key=len)
print(unique_names)

file = open("processed_names.txt", "w")
```

```
for names in unique_names:
    file.write(name)
```

```
['OM', 'Rohan', 'Aayush', 'Gaurav', 'Prachi', 'Prasad', 'Pranav', 'Rudraksh', 'Sh
rushti']
```

In [66]:
```
#10.
groups = defaultdict(list)
for name in names:
    groups[name[0].upper()].append(name)
for letter, group in groups.items():
    with open(f"{letter}_names.txt", "w") as file:
        for name in group:
            file.write(name)
```

In [ ]:
```

```